



CHANDRA SHEKHAR AZAD PARK

SYSTEM ARCHITECTURE & END-TO-END PROJECT FLOW SPECIFICATION

Version: 2.0 (Production)

Platform: Web / PWA /
Serverless

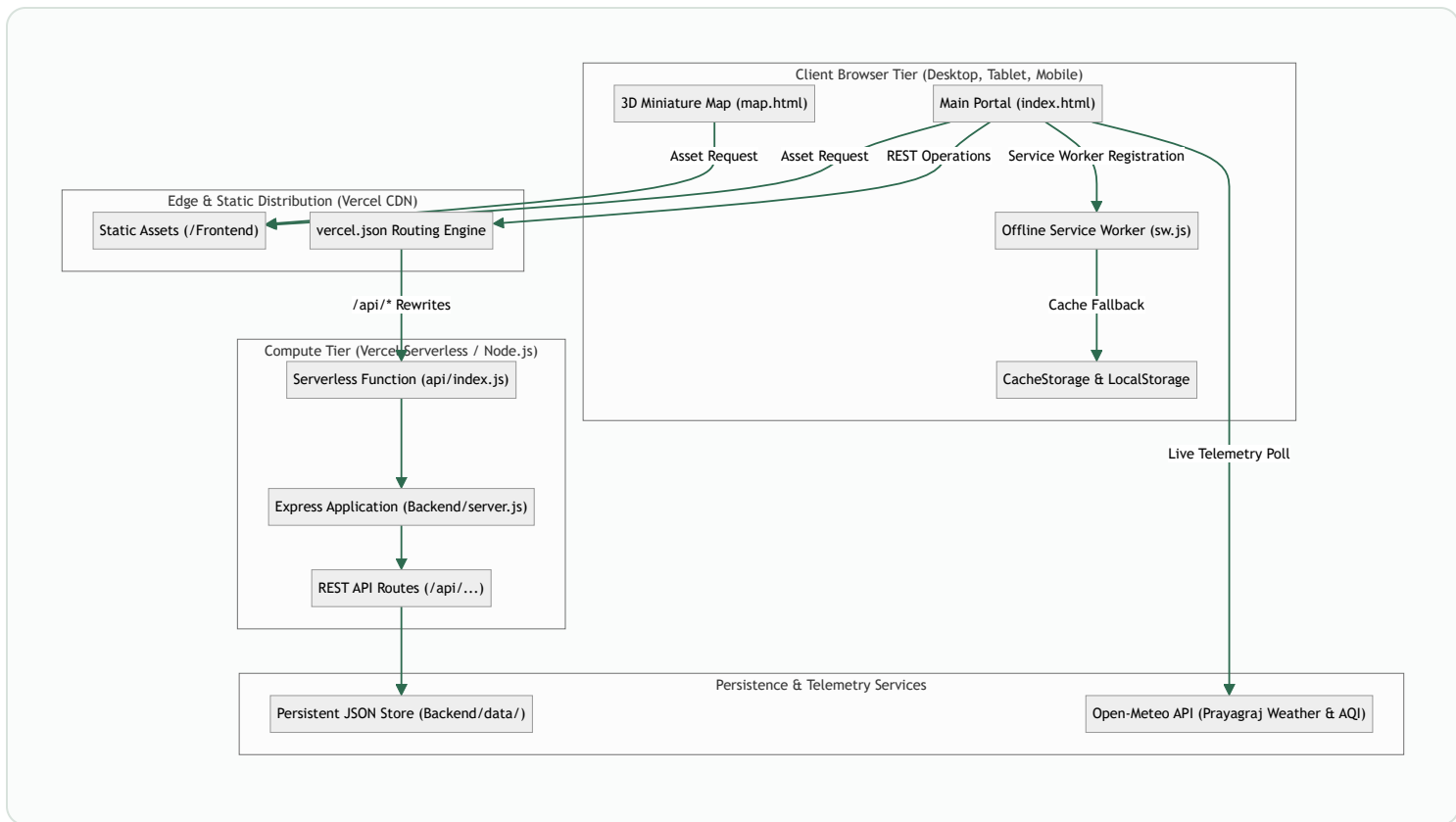
Developer: Abhishek
(Abhitech8code)

Generated: October 2026

1. EXECUTIVE SUMMARY & SOLUTION ARCHITECTURE

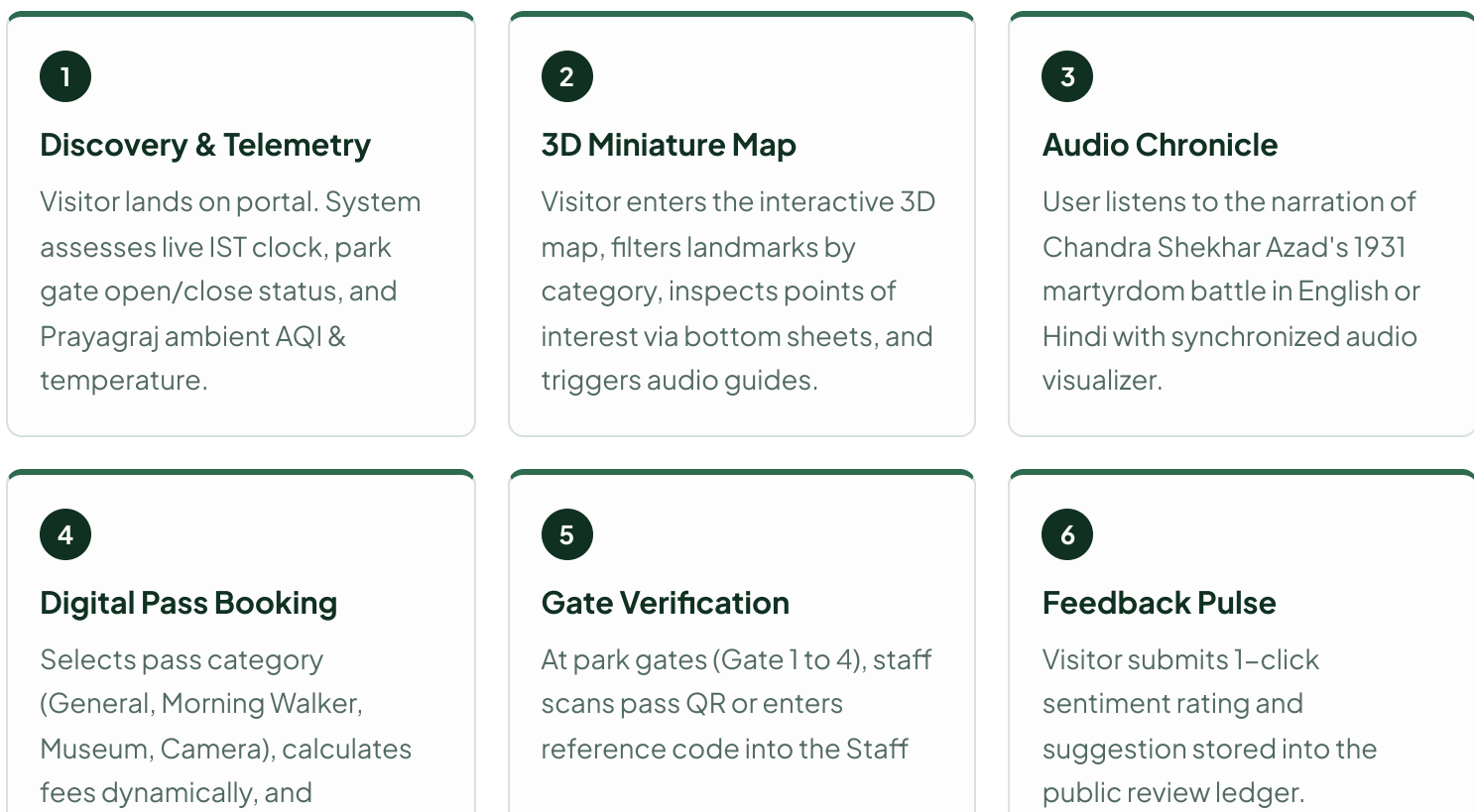
The **Chandra Shekhar Azad Park Digital Heritage & Visitor Platform** is an independent, production-grade tourism and cultural preservation portal for the historic 133-acre sanctuary in Prayagraj, Uttar Pradesh. It combines real-time ambient telemetry, interactive isometric 3D mapping, dual-language audio chronicles, digital pass issuance with cryptographic QR tokens, and gatekeeper verification desks.

Core Architectural Philosophy: Zero external frameworks (Vanilla HTML/CSS/ES6+), extreme mobile responsiveness (zero horizontal scroll down to 320px), client-side offline resilience with Service Workers, and low-latency serverless API execution on Vercel Edge networks.



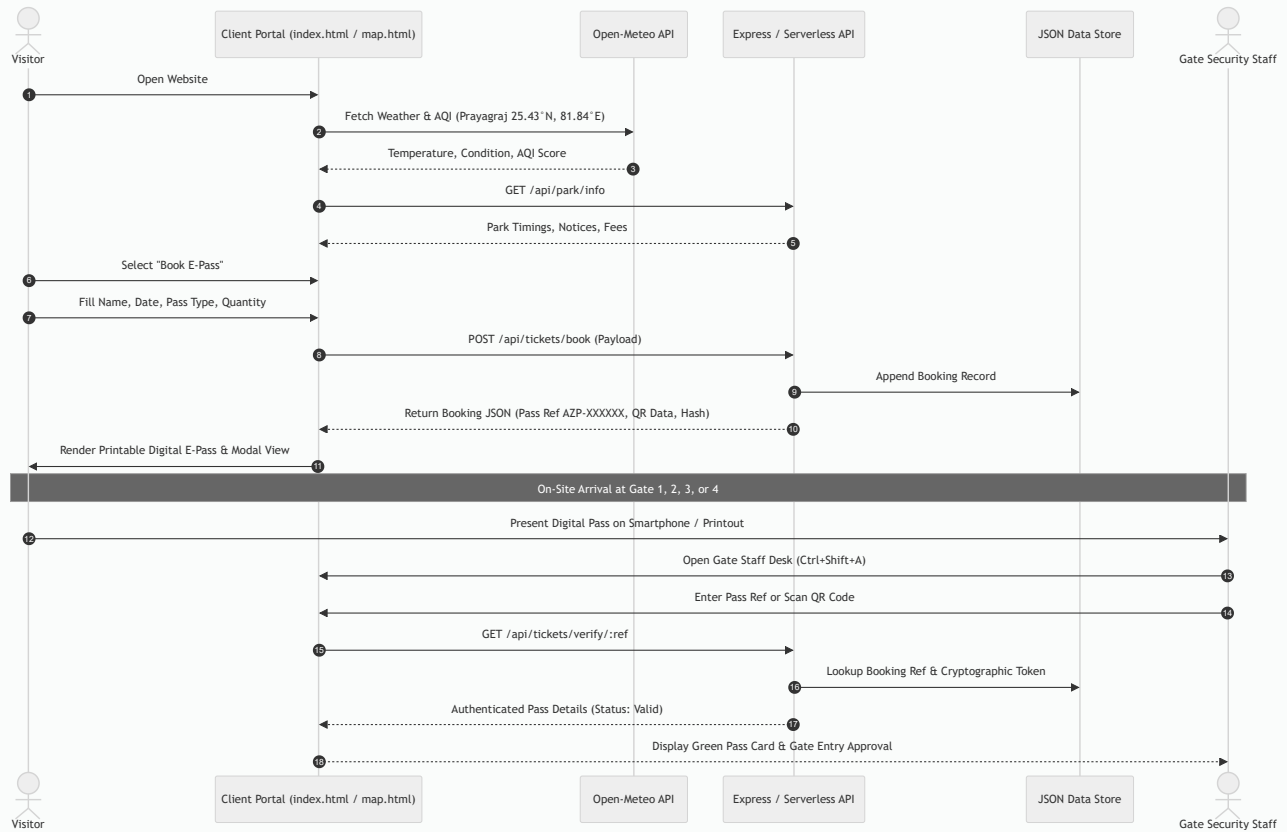
2. COMPLETE END-TO-END USER FLOW (VISITOR JOURNEY)

The platform facilitates a seamless end-to-end journey from online pre-visit planning to on-site exploration and post-visit feedback.



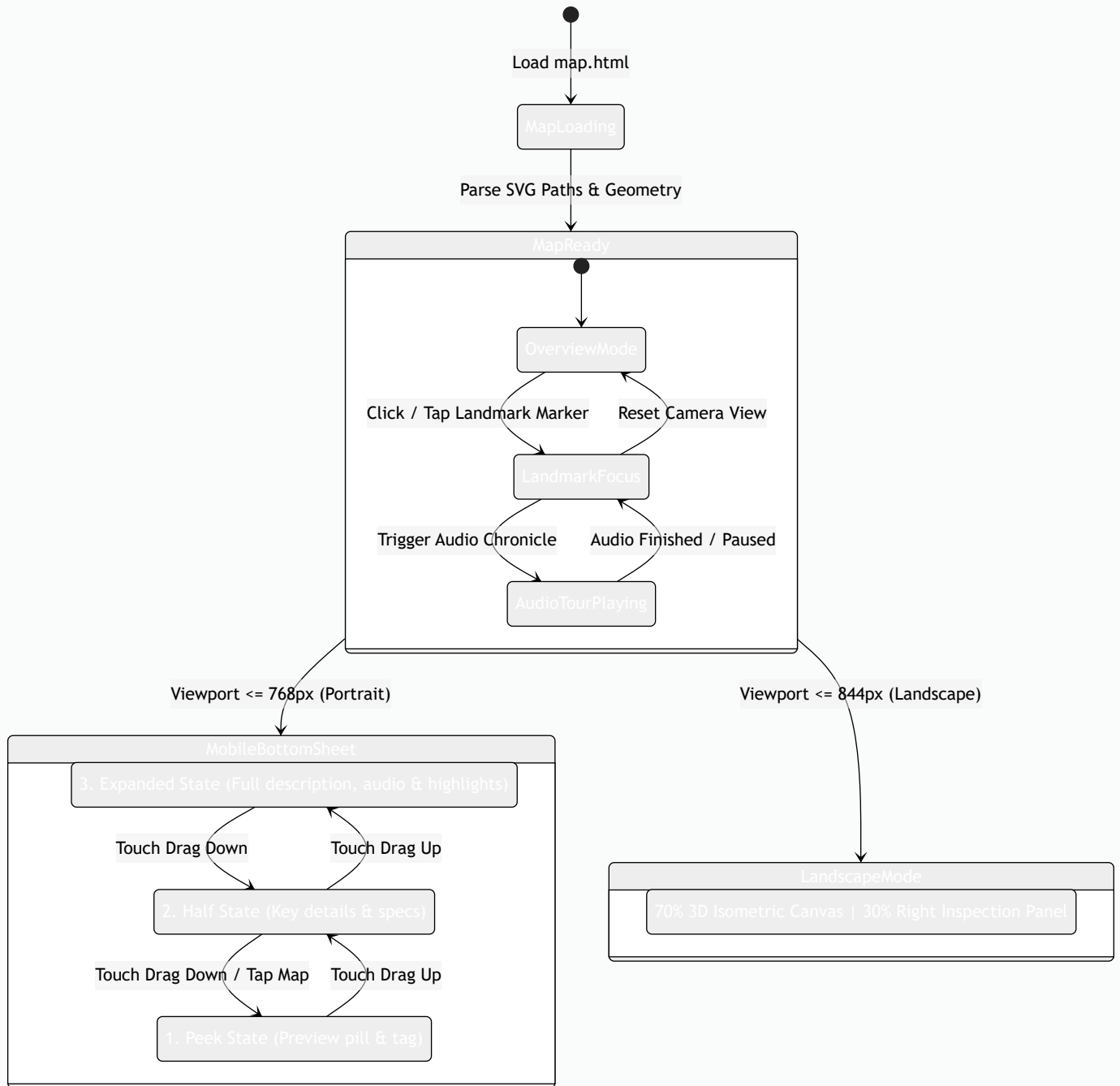
generates verified E-Pass with QR code.

Desk modal for cryptographic authentication.



3. SIGNATURE 3D INTERACTIVE MAP FLOW

The 3D interactive map (`Frontend/map.html`) serves as a digital miniature model of the park. It incorporates responsive layout adaptation, touch gestures, dynamic camera framing, and audio tour coordination.

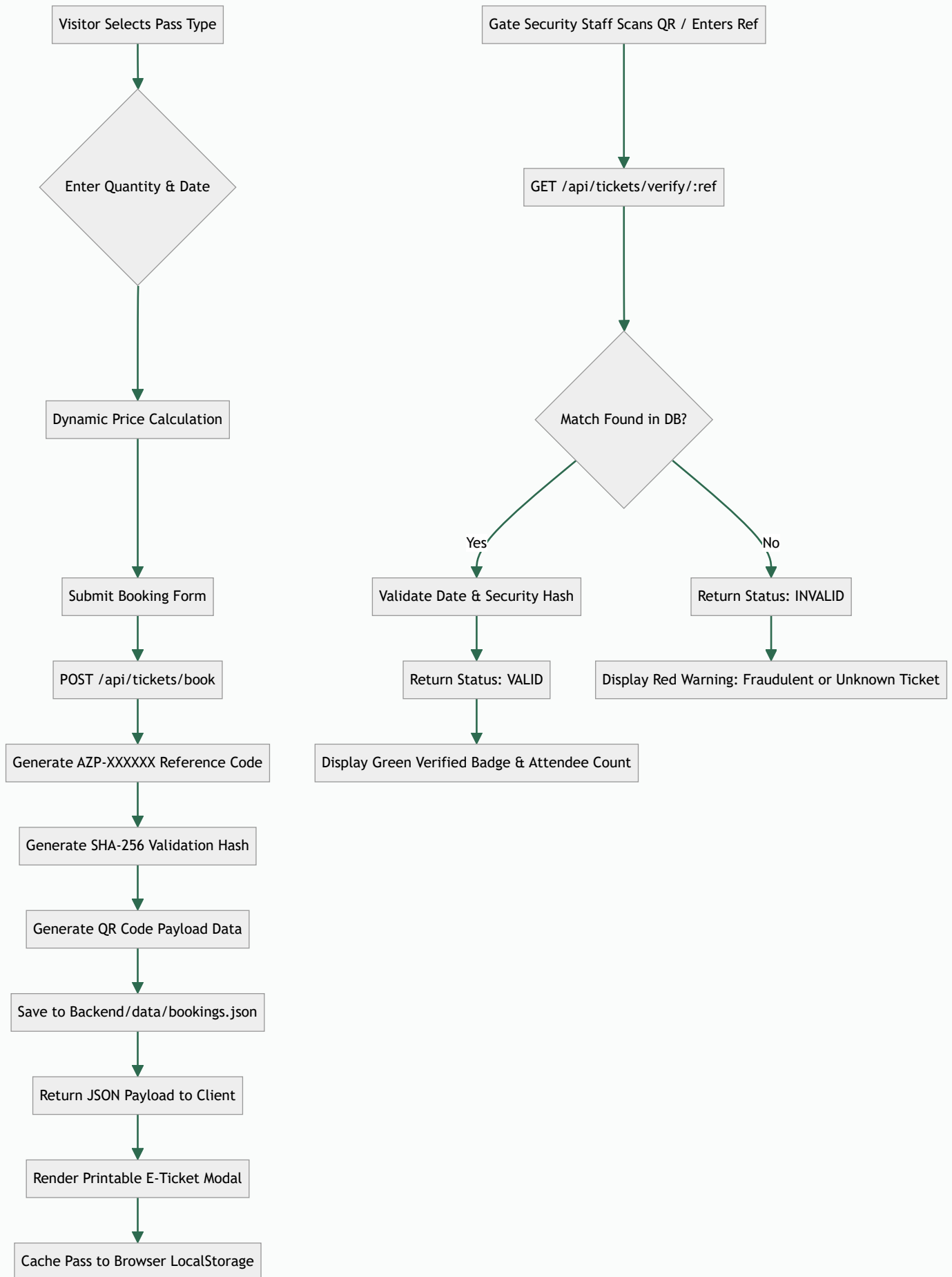


3.1 Map Technical Features

Feature	Mechanism	User Benefit
Isometric Projection	Vector SVG styling with depth offsets, drop shadows, and 3D path transforms	Digital miniature look without high GPU/WebGL memory overhead
Radar Tracking	Proportional landmark pin layout overlaying miniature sanctuary radar	Instant geographical orientation of the 133-acre sanctuary
Multi-Category Filters	Real-time DOM attribute filtering (All, Heritage, Museums, Library, Flora, Promenades)	Instant discovery of specific heritage interests without page reloads
Audio Chronicle Integration	HTML5 Web Audio API with synchronized Hindi / English historical narration	Immersive museum-grade audio tour while exploring landmarks

4. PASS BOOKING & VERIFICATION PIPELINE

The ticketing engine generates verifiable digital passes with unique reference identifiers and verification tokens.



4.1 Ticket Pricing Structure

Pass Tier	Tariff	Validity	Privileges
General Day Pass	₹15 / adult (Children < 5 Free)	Single Day Entry	Access to all lawns, trails, memorials, and fountain area
Morning Walker Monthly	₹120 / month	30 Days	Exclusive 05:00 AM – 08:30 AM fitness access across all gates
Morning Walker Annual	₹1,000 / year	365 Days	Year-round unlimited health trail and jogging track entry
Museum Combined Pass	₹50 / visitor	Single Day	Park entry + Allahabad National Museum galleries entry
Professional Camera Permit	₹100 / camera	Single Day	DSLR / Mirrorless photography authorization

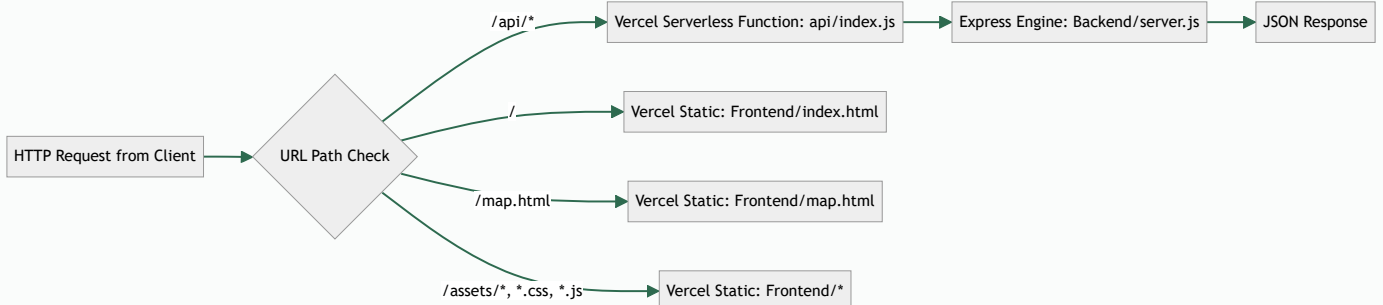
5. REST API ARCHITECTURE & ENDPOINTS

The server exposes lightweight JSON endpoints designed for instant execution on Node.js and Vercel Serverless Functions.

Endpoint	Method	Parameters	Description
<code>/api/park/info</code>	GET	None	Retrieves metadata, opening hours, gate notices, and emergency contacts
<code>/api/attractions</code>	GET	<code>?category=...</code>	Lists all monuments, pavilions, library, museum, and coordinates
<code>/api/events</code>	GET	None	Returns community yoga, heritage walks, and horticultural exhibitions
<code>/api/reviews</code>	GET / POST	JSON: author, rating, city, text	Reads visitor guestbook and accepts verified guestbook reviews
<code>/api/feedback</code>	GET / POST	JSON: sentiment, suggestion, category	Aggregates visitor sentiment metrics (great, good, okay, needs-improvement)
<code>/api/tickets/book</code>	POST	JSON: name, email, date, ticketType, quantity	Issues verifiable digital E-Pass with reference number and QR payload
<code>/api/tickets/verify/:ref</code>	GET	Path: <code>ref</code>	Security verification endpoint validating ticket authenticity and entry status
<code>/api/health</code>	GET	None	System health, uptime, and server timestamp check

6. VERCEL CLOUD ROUTING & EDGE DEPLOYMENT

The project is structured for zero-configuration, continuous Git-integrated deployment via `vercel.json`.



```
{
  "version": 2,
  "rewrites": [
    { "source": "/api/(.*)", "destination": "/api/index.js" },
    { "source": "/", "destination": "/Frontend/index.html" },
    { "source": "/map.html", "destination": "/Frontend/map.html" },
    { "source": "/(.*)", "destination": "/Frontend/$1" }
  ]
}
```

7. SECURITY, PRIVACY & PERFORMANCE AUDITS

- **Zero Secrets in Version Control:** Audited `.gitignore` preventing tracking of `.env`, private keys, database URLs, deployment tokens, and temporary files.
- **Input Sanitization:** Form inputs for bookings, reviews, and pulse suggestions are sanitized against XSS attacks.
- **Responsive Resilience:** Verified zero horizontal canvas overflow across all standard mobile and desktop viewports (320px, 375px, 390px, 768px, 1024px, 1440px).
- **Offline Survivability:** Service Worker caches essential assets for offline reading and wayfinding.